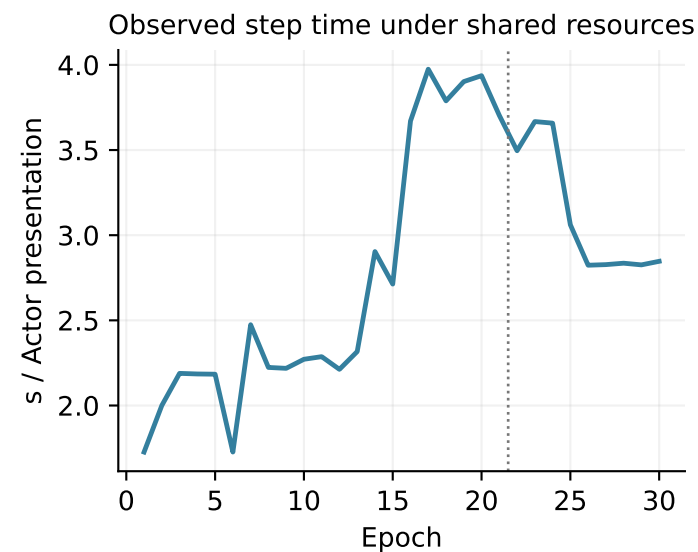
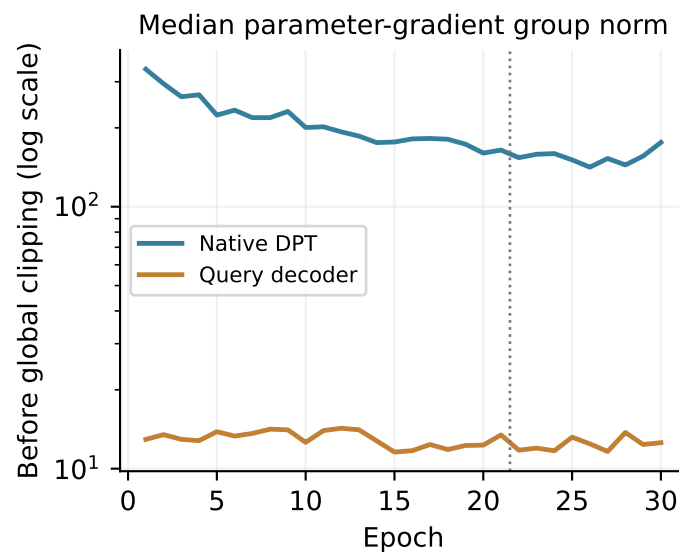
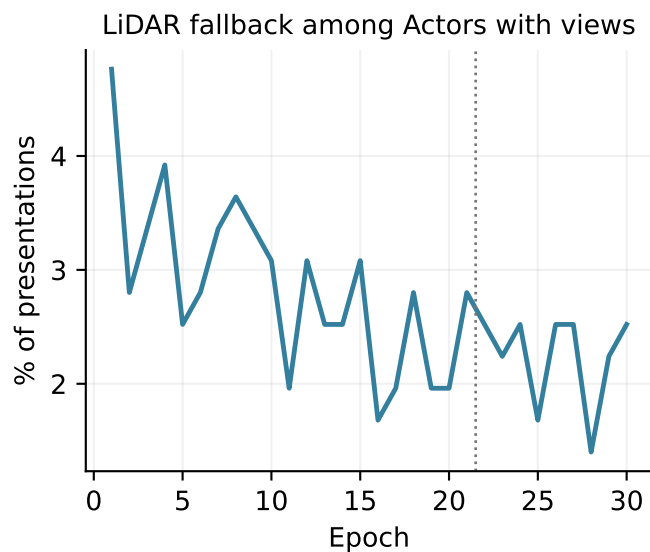
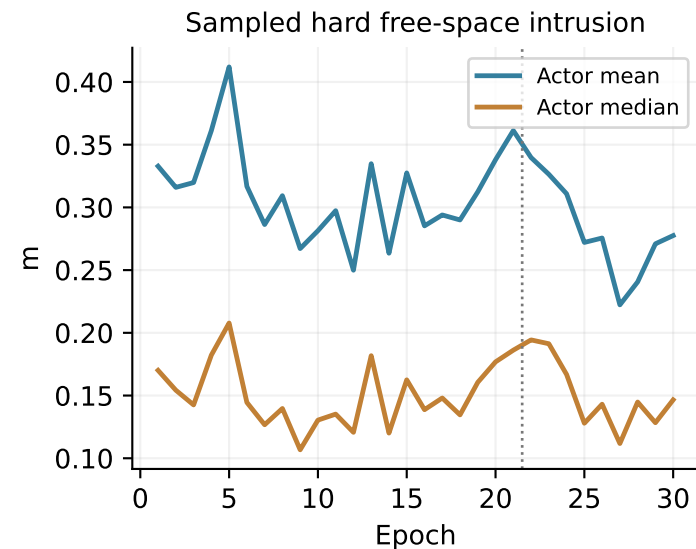
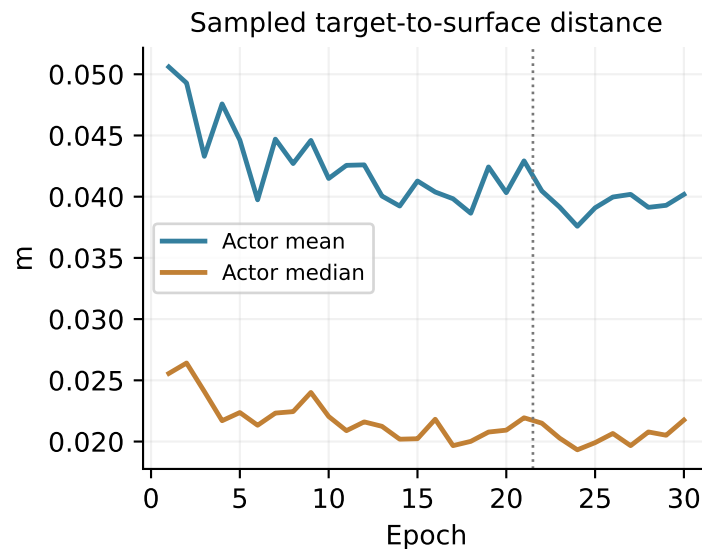
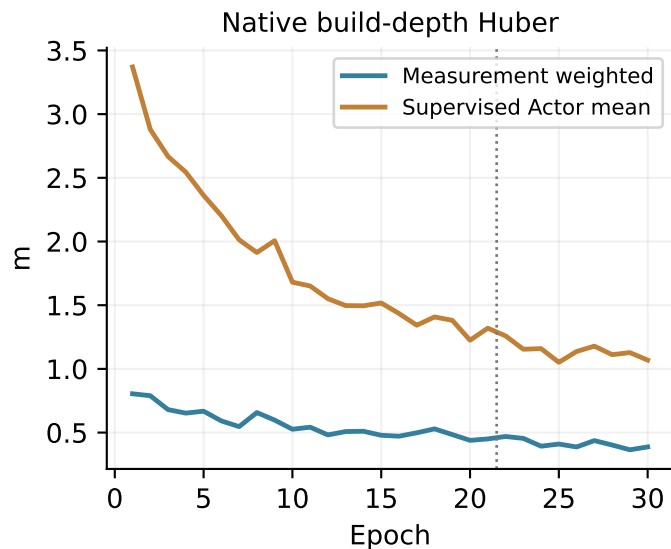


Joint native geometry + spatial queries: training process

371 fit Actors per complete epoch; sampled targets and rays vary, so this is not paired generalization evidence.



Dotted line: recovery from epoch 21. Saved complete epochs are counted once; 107 discarded parent updates remain separately recorded.
Native loss excludes unobserved Actors. Memory contention and concurrent work affect timing; loss reduction does not establish physical accuracy.